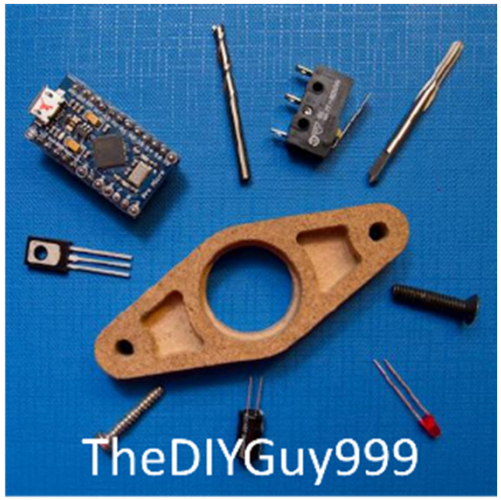
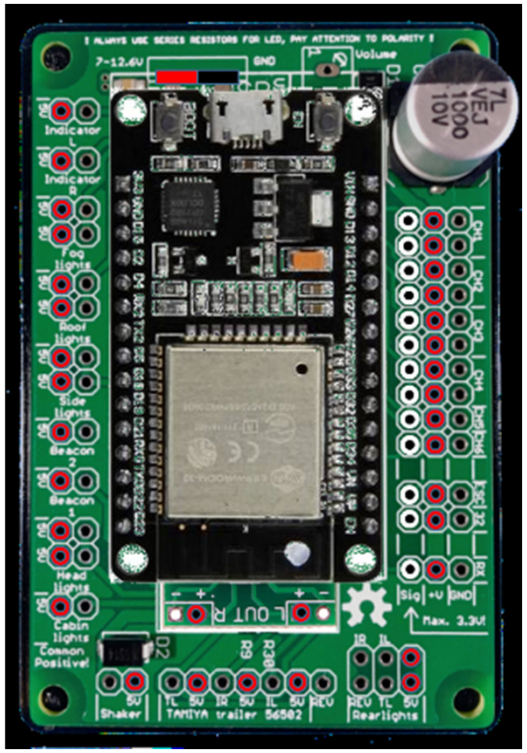
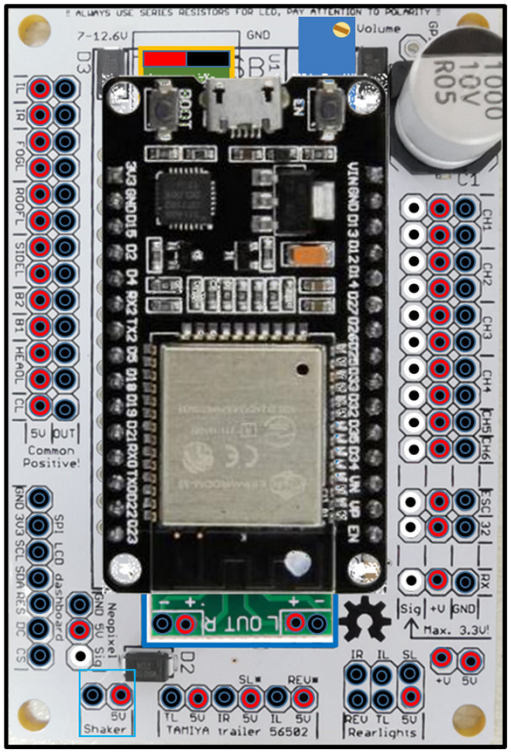
ESP32 RC Sound & Light Controller

Board installation and Vehicle set-up







INTERNAL

ESP32 RC Sound & Light Controller

Safety Regulations

This product must not be used until this manual has been fully read and understood. Likewise, you must agree with all the restrictions mentioned in this section.

- This open source product is left to the user's own responsibility for use, free development and customization. TheDIYGuy999 assumes no liability whatsoever for any damage or consequential damage resulting from the use of this system.
- This product is not suitable for large and dangerous models, because there is no redundancy.
- Never use this product on public roads.
- This product has not been tested for FCC or CE compliance.
- Never connect or disconnect wires while the product is connected to a battery.
- **Always disconnect the battery, when this product is not in use. It always draws a couple of Milliamps, even if the ESC is switched off and will drain the battery otherwise!**
- This product is not protected against wrong polarity! Always double check your wiring before connecting the battery. Wrong polarity will immediately destroy the product.
- Always insulate the vehicle wiring properly, using heat shrinks in order to prevent short circuits. Never mount the bare PCB directly on top of a metal plate. This will cause short circuits.
- This product may only be used in dry locations.
- This product is not suitable for children below 14 years.

Source: TheDIYGUY999 Quick start manual

INTERNAL

ESP32 RC Sound & Light Controller

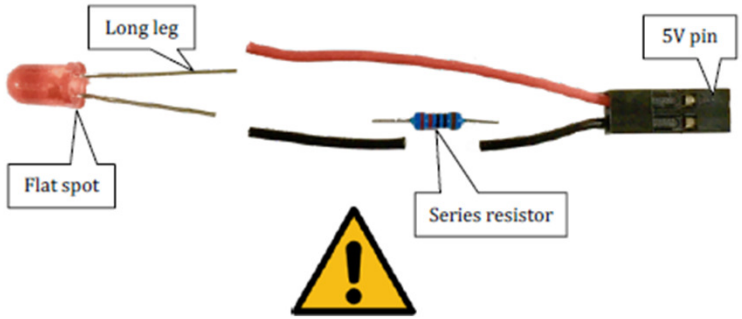
LED Wiring

The LED supply voltage is **5V**, coming from the internal regulator. A common LED maximum current is **15mA**.
The following chart provides the required minimum series resistor Values.

LED color	Forward voltage [Volts]	Minimum resistor [Ohms]
Red	1.8	220
Orange	1.9	220
Green	2	200
White	3	150
Blue	3	150

- You can also use a calculator in order to determine the correct value:

https://www.digikey.de/en/resources/conversion-calculators/conversion-calculator-led-series-resistor



Source: TheDIYGUY999 Quick start manual

INTERNAL

ESP32 RC Sound & Light Controller

LED Wiring

- The LED will **not light up**, if connected the **wrong way** around.
- **Never connect** LED or other devices **without** a suitable **series resistor** to the LED output pins (except the TAMIYA trailer connector). This will **permanently damage** the integrated ULN2003 LED drivers.
- If more than one LED is wired in parallel to an LED output pin, each one needs its own series resistor.
- Two red or orange LED can be wired in series. This makes the circuit **more efficient**. In this case you need to add both series voltages together for the series resistor calculation.
- The **maximum average LED output current per channel is 100mA**. However, it is not recommended to load each channel with this current. Otherwise, the ULN2003 LED driver may **overheat**. Always **check it** for excessive temperature raise for at least 10 minutes after you wired a new vehicle. If the temperature is too high to the touch, you should use an external driver transistor for the LED.
- Also note, that the **maximum sum current** for the entire 5V rail, which is supplied by the on-board regulator is **1A**.

Source: TheDIYGUY999 Quick start manual

INTERNAL

ESP32 RC Sound & Light Controller

Board Overview

Lights

Indicator L

Indicator R

Fog light

Roof light*

Side light

Beacon 2

Beacon 1

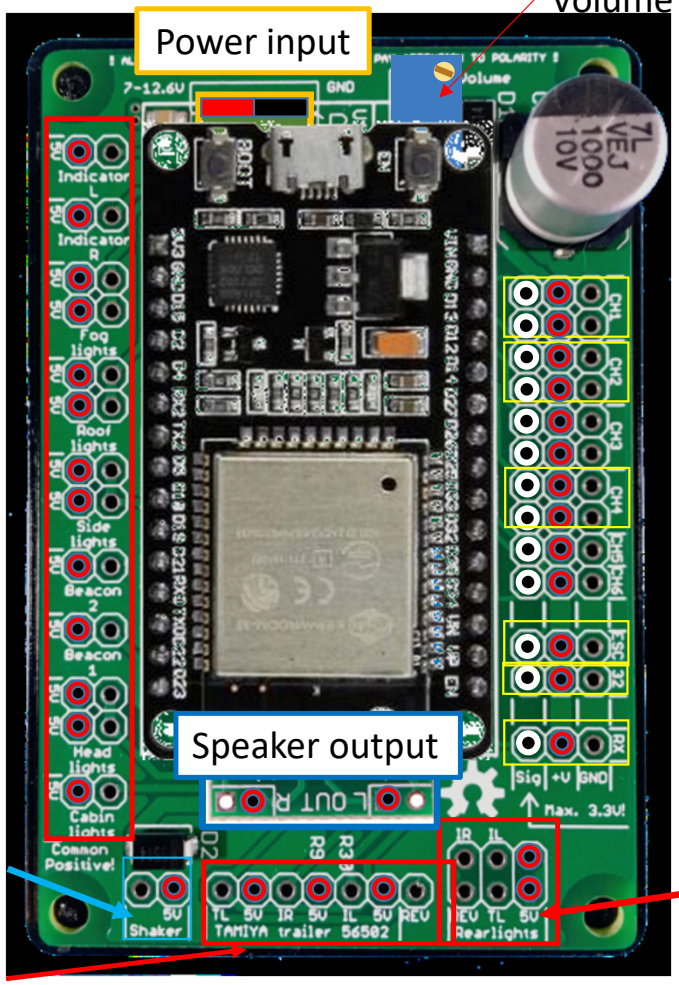
Head light

Cabin light

Shaker Output

TAMIYA Trailer LED connector

Roof light* - In case of #Separate_headlights plug main beams into Roof lights Plug Roof lights into Side lights



Steering

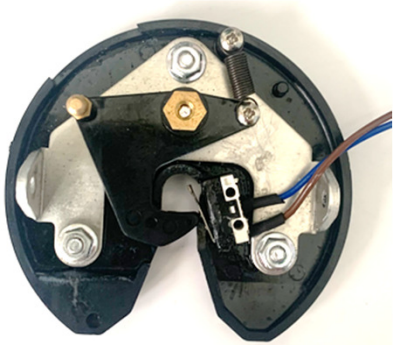
Gearbox

5th Wheel servo

HW1080 ESC

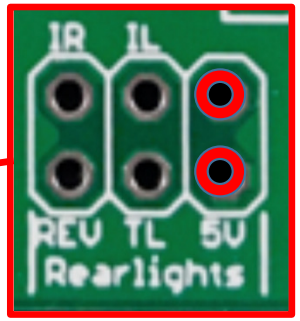
Trailer presence switch

RX Input



Limit switch on 5th Wheel (coupler) to detect trailer king pin.

Rear lights



INTERNAL

ESP32 RC Sound & Light Controller

open source hardware

Board Overview

Power input 2S/3S

Volume control

Lights

Indicator L
Indicator R

Fog light

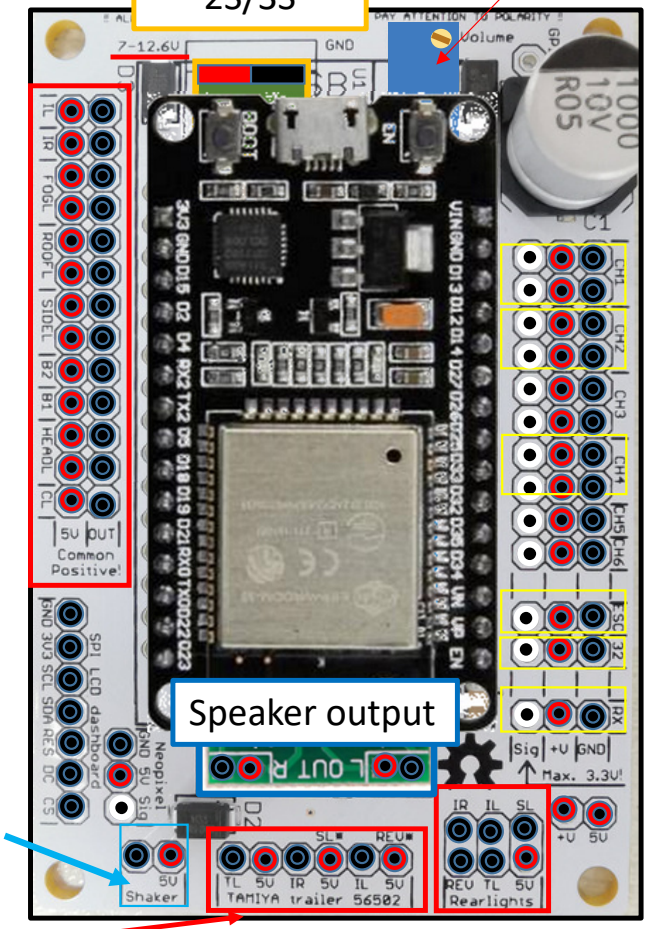
Roof light*

Side light

Beacon 2
Beacon 1

Head light

Cabin light



Speaker output

Steering

Gearbox

5th Wheel servo

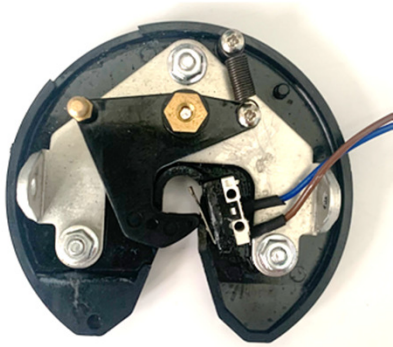
HW1080 ESC
Trailer presence switch

RX Input

Shaker Output

TAMIYA Trailer LED connector

Roof light* - In case of #Separate_headlights plug main beams into Roof lights Plug Roof lights into Side lights. Please see next slide



Limit switch on 5th Wheel (coupler) to detect trailer king pin.

ESP32 RC Sound & Light Controller

6_Lights.h

src > C 6_Lights.h > ...

```
// These light options are adjustable on the configuration website and stored in the EEPROM ----------------------------------------
bool noCabLights = false;                 // The cablights step in the lights sequence is skipped, if true
bool noFogLights = false;                 // The foglights step in the lights sequence is skipped, if true
bool xenonLights = false;                 // Headlights will show a xenon bulb ignition flash, if true
bool flickeringWileCranking = false;       // if true: The lights will flicker a bit during engine cranking, otherwise they are just a bi
bool ledIndicators = false;               // indicators are switching "hard" (without fading like an incandescent bulb), if true (was LED
bool swap_L_R_indicators = false;         // If you want to swap L & R indicators
bool indicatorsAsSidemarkers = false;      // "US Mode" The indicators are used as side markers as well. This is commonly used in US truc
bool separateFullBeam = false;            // The full beam is a separate bulb, connected to roof lights pin. Connect roof lights to side
bool flashingBlueLight = true;            // double flash blue lights if "true", "rotating" beacons if "false"
bool hazardsWhile5thWheelUnlocked = true; // Hazards on, if 5th wheel is unlocked
```

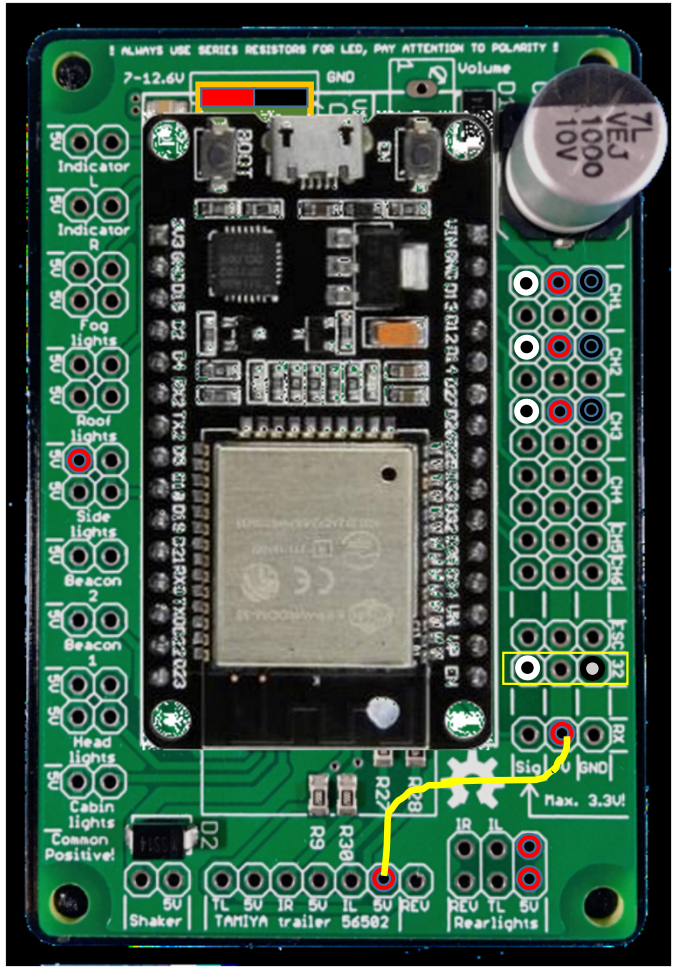
INTERNAL

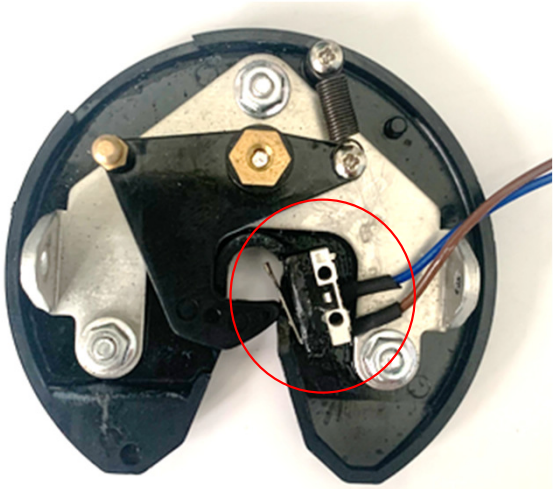
```
// Flysky FS-i6S remote configuration profile (CAT 730) ----------------------------------------------------------
#ifdef FLYSKY_FS_I6S

// Channel assignment (use NONE for non existing channels!)
// Remote channel ######   // Sound controller channel ##########################################
#define STEERING 1            // CH1 steering
#define GEARBOX 10   SwC      // CH2 3 position switch for gearbox (left throttle in tracked mode)
#define THROTTLE 3            // CH3 throttle & brake (right throttle in tracked mode)
#define HORN 6       VrB      // CH4 horn and bluelight / siren
#define FUNCTION_R 2          // CH5 jake brake, high / low beam, headlight flasher, engine on / off
#define FUNCTION_L 4          // CH6 indicators, hazards
#define POT2 7       SwD      // CH7 pot 2 dump bed
#define MODE1 8      SwB      // CH8 mode 1 switch
#define MODE2 NONE            // CH9 mode 2 switch
#define MOMENTARY1 NONE       // CH10
#define HAZARDS NONE          // CH11
#define INDICATOR_LEFT NONE   // CH12
#define INDICATOR_RIGHT NONE  // CH13
```

INTERNAL

ESP32 RC Sound & Light Controller

Trailer card





Limit switch on 5th Wheel (coupler) to detect trailer king pin.

Optional Trailer presence switch between Sig and Gnd if not fitted to 5th wheel. If no switch, bridge out.

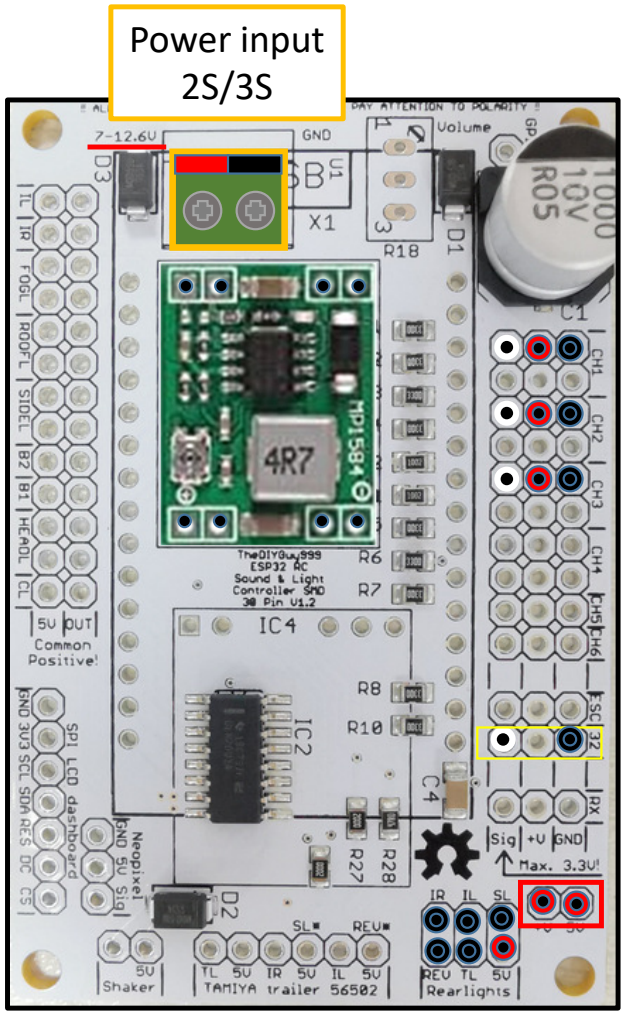
Connect power wire to headers if no ESC is used.
Unplug lead if ESC is used.

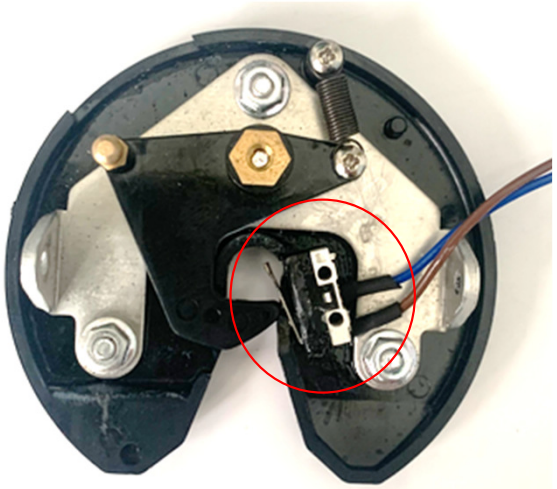
INTERNAL

ESP32 RC Sound & Light Controller

Trailer card

open source hardware





Limit switch on 5th Wheel (coupler) to detect trailer king pin.

Optional Trailer presence switch between Sig and Gnd if not fitted to 5th wheel. If no switch, bridge out.

Connect power jumper to headers if no ESC is used.
Unplug jumper if ESC is used.

INTERNAL

ESP32 RC Sound & Light Controller

Hobbywing 1080 ESC setup

Set up your HW1080 prior to connecting to the Sound Card.
Ensure the throttle is calibrated. (plug ESC into RX Ch3 and set points)

https://youtu.be/RRYwr14FwjE

INTERNAL

ESP32 RC Sound & Light Controller

Hobbywing 1080 ESC setup

Program the ESC with the program card

Item:	Value
1:	3
2:	1 (1= LiPo, 2 = NIMH battery)
3:	3
4:	3
5:	4
6:	2
7:	9
8:	1
9:	8 (change it, important)
10:	4
11:	4
12:	5
13:	5 (16KHz = less whining)
14:	1 (be careful here, this will change the BEC voltage!!)
15:	1 (change it, important)

INTERNAL

ESP32 RC Sound & Light Controller

Hobbywing 1080 ESC setup

open source hardware

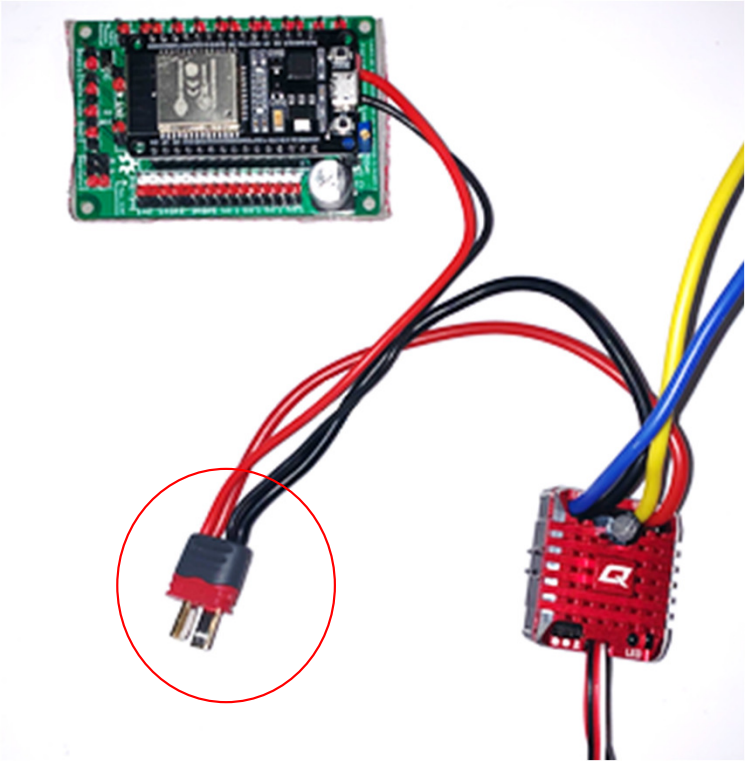
Programmable Item	Option 1	Option 2	Option 3	Option 4	Option 5	Option 6	Option 7	Option 8	Option 9
1.Running Mode	Fwd/Brk	Fwd/Rev/Brk	Fwd/Rev						
2.Battery Type	LiPo	NiMH							
3.Cutoff Voltage	Disabled	Auto (Low)	Auto (Medium)	Auto (High)					
4. Initial Start Force	0%	2%	4%	6%	8%	10%	12%	14%	16%
5.Max. Forward Force	25%	50%	75%	100%					
6.Max.Reverse Force	25%	50%	75%	100%					
7. Max. Brake Force	0%	12.5%	25%	37.5%	50%	62.5%	75%	87.5%	100%
8.Initial Brake Force	0%	6.25%	12.5%	18.75%	25%	31.25%	37.5%	43.75%	50%
*9. Drag Brake	0%	5%	10%	50%	60%	70%	80%	90%	100%
10.Drag Brake Rate	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	Level 9
11.Neutral Range	0.02ms	0.03ms	0.04ms	0.05ms	0.06ms	0.07ms	0.08ms	0.10ms	0.12ms
12. Start Mode/Punch	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	Level 9
13.PWM Frequency	1K	2K	4K	8K	16K				
*14.BEC Voltage	6V	7.4V							
*15.Freewheeling	Enabled	Disabled							

INTERNAL

ESP32 RC Sound & Light Controller

Battery connections – Option 1

1. I power my board by soldering the wires to the ESC plug
2. I would suggest putting a switch on the wires leading to the board…or disconnect from battery when not in use



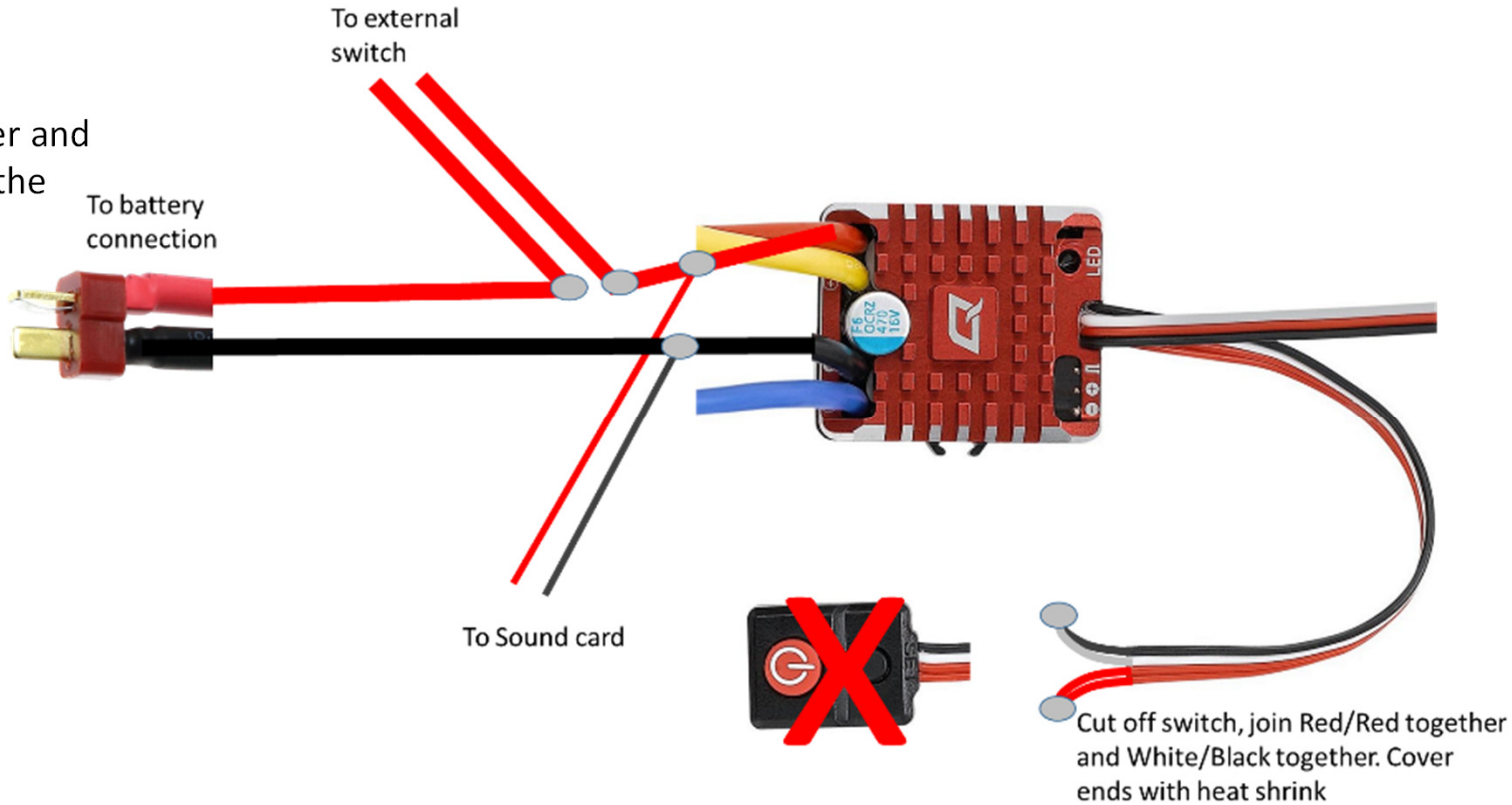
- **Always disconnect the battery, when this product is not in use. It always draws a couple of Milliamps, even if the ESC is switched off and will drain the battery otherwise!**

INTERNAL

ESP32 RC Sound & Light Controller

Battery connections – Option 2 (ensure all throttle calibrations has been done)

1. To have only one external switch…
2. I did this on my King Hauler and will convert the Scania to the same.



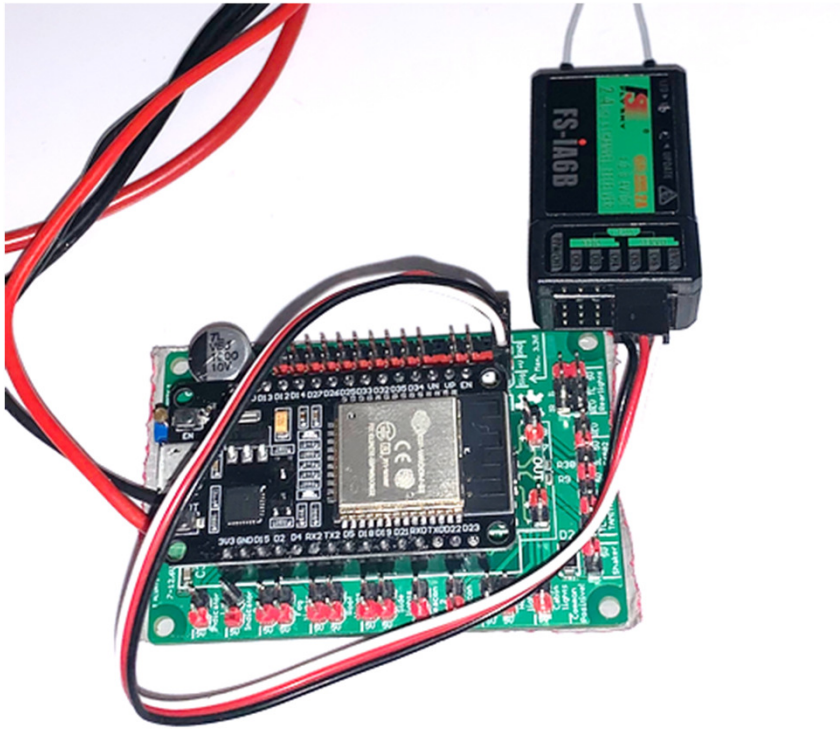
- **Always disconnect the battery, when this product is not in use. It always draws a couple of Milliamps, even if the ESC is switched off and will drain the battery otherwise!**

INTERNAL

ESP32 RC Sound & Light Controller

Receiver connection

1. If IBUS or SBUS communication is selected, only one wire is needed from the receiver to the Sound Card



INTERNAL

ESP32 RC Sound & Light Controller

Flysky FS-i6X modified setup to free up Pot A “VrA” to Ch6 - control other electronic function

ESP32 sound controller						
PWM	PPM	SBUS	IBUS	Type	Sound controller channel	Name
Yes, CH1 - CH4, 35, PPM *	Yes, RX	Yes, SBUS	Yes, RX	analog or 3 position	1	STEERING
					2	GEARBOX
					3	THROTTLE
					4	HORN
					5	FUNCTION_R
					6	FUNCTION_L
No					7	POT A
				latching 2 position	8	MODE1
	No				9	MODE2
				momentary	10	MOMENTARY1
				flags	11	HAZARDS
					12	INDICATOR_LEFT
					13	INDICATOR_RIGHT

FLYSKY FS-i6X (with lightbar)					
Joystick, button, potentiometer				Function	
Remote channel	Name	Direction	Duration	Switch A up (100% DR)	Switch A down (75% DR)
1	Stick horizontal right			Steering	
8	Switch C (3 pos.)			Gearbox shifting	
3	Stick vertical left			Throttle & brake	
5	Pot B "VrB"			Horn, bluelight & siren	
2	Stick vertical right	up	togggle long	High / low beam	light state up
			toggle		
			momentary	Headlights flasher	
		down	togggle long	volume (1	Engine on / off
			toggle		
			momentary	Jakebrake	
4	Stick horizontal left	left	togggle long		hazards
			toggle	left indicator on / off	
			momentary		
		right	togggle long		5th wheel unlocking (3
			toggle	right indicator on / off	
			momentary		
6	Pot A "VrA"			Lightbar (connect to CH6 on Flysky RX)	
7	Switch D			Transmission neutral (2	
9	Switch B			Sound 1	
16				not available (Switch A internally used for DR)	
16				not available	
16				not available	
16				not available	

INTERNAL

ESP32 RC Sound & Light Controller

Flysky FS-i6X Standard setup

ESP32 sound controller							FLYSKY FS-i6X					
Usable with, Header:				Type	Sound controller channel	Name	Joystick, button, potentiometer				Function	
PWM	PPM	SBUS	IBUS				Remote channel	Name	Direction	Duration	Switch A up (100% DR)	Switch A down (75% DR)
Yes, CH1 - CH4, 35, PPM *	Yes, RX	Yes, SBUS	Yes, RX	analog or 3 position	1	STEERING	1	Stick horizontal right			Steering	
					2	GEARBOX	6	Switch C (3 pos.)			Gearbox shifting	
					3	THROTTLE	3	Stick vertical left			Throttle & brake	
					4	HORN	5	Pot B "VrB"			Horn, bluelight & siren	
					5	FUNCTION_R	2	Stick vertical right	up	togggle long	High / low beam	light state up
										toggle		
										momentary	Headlights flasher	
									down	togggle long	volume (1	Engine on / off
										toggle		
										momentary	Jakebrake	
					6	FUNCTION_L	4	Stick horizontal left	left	togggle long		hazards
										toggle	left indicator on / off	
										momentary		
									right	togggle long		5th wheel unlocking (3
										toggle	right indicator on / off	
										momentary		
No					7	POT2	8	Pot A "VrA"			not used	
				latching 2 position	8	MODE1	7	Switch D			Transmission neutral (2	
	No				9	MODE2	9	Switch B			Sound 1	
				momentary	10	MOMENTARY1	16				not available (Switch A internally used for DR)	
				flags	11	HAZARDS	16				not available	
					12	INDICATOR_LEFT	16				not available	
					13	INDICATOR_RIGHT	16				not available	

INTERNAL

ESP32 RC Sound & Light Controller

Flysky FS-i6X modified setup to use Pot A “VrA” to control “other”

Modified code on Remote Tab

```
// Channel assignment (use NONE for non existing channels!)
// Remote channel #######   // Sound controller channel ##################################
#define STEERING 1           // CH1 steering
//#define GEARBOX 6          // CH2 3 position switch for gearbox (left throttle in tracked mode)
#define GEARBOX 8            // CH2 3 position switch for gearbox (left throttle in tracked mode)
#define THROTTLE 3           // CH3 throttle & brake (right throttle in tracked mode)
#define HORN 5               // CH4 horn and bluelight / siren
#define FUNCTION_R 2         // CH5 jake brake, high / low beam, headlight flasher, engine on / off
#define FUNCTION_L 4         // CH6 indicators, hazards
//#define POT2 8             // CH7 pot 2
#define POT2 6               // CH7 pot 2
#define MODE1 7              // CH8 mode 1 switch
#define MODE2 9              // CH9 mode 2 switch
#define MOMENTARY1 NONE          // CH10
#define HAZARDS NONE             // CH11
#define INDICATOR_LEFT NONE      // CH12
#define INDICATOR_RIGHT NONE     // CH13
```

INTERNAL

ESP32 RC Sound & Light Controller

Flysky FS-i6X Dual rate setup

≡ FUNCTIONS ≡
- Throttle curve
- Mixes
- V Tail
- ➡ Assign switches
- Throttle hold

Switch "SWA" up

: Assign switches :

➡ Fly mode	SwA
Normal	
Idle Mode	None
Thro. hold	None

Switch "SWA" down

: Assign switches :

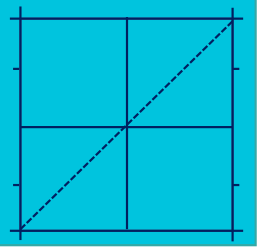
➡ Fly mode	SwA
Sport	
Idle Mode	None
Thro. hold	None

≡ FUNCTIONS ≡
- End points
- Display
- Aux. channels
- Subtrim
- ➡ Dual rate/exp.

Switch "SWA" up

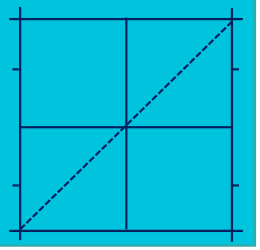
= Dual rate / exp . =

Normal
Ch2
➡ Rate 100
Exp 0



= Dual rate / exp . =

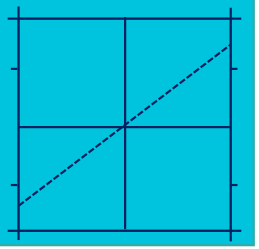
Normal
Ch4
➡ Rate 100
Exp 0



Switch "SWA" down

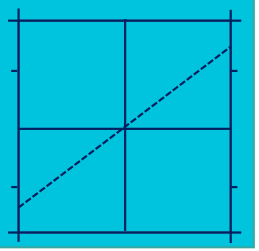
= Dual rate / exp . =

Sport
Ch2
➡ Rate 75
Exp 0



= Dual rate / exp . =

Sport
Ch4
➡ Rate 75
Exp 0



INTERNAL

ESP32 RC Sound & Light Controller

Flysky FS-i6X standard setup

open source hardware

```
≡ FUNCTIONS ≡
End points
Display
➡ Aux. channels
Subtrim
Dual rate/exp.
```

```
═ Aux . Channels ═
Channel 5
Source   VrB

Channel 8
Source   VrA
```

Horn/Beacon lights / trailer control

External – plug into Rx Ch 6 (e.g. control light switch)

```
═ Aux . Channels ═
Channel 7
Source   SwD

Channel 6
Source   SwC
```

Virtual neutral (rev truck)

3 position Gear shift

```
═ Aux . Channels ═
Channel 9
Source   SwB

Channel 10
Source   None
```

Un-couple 5th wheel servo (trailer control)

INTERNAL

ESP32 RC Sound & Light Controller

Flysky FS-i6X modified setup

open source hardware

FUNCTIONS

- End points
- Display
- ➡ Aux. channels
- Subtrim
- Dual rate/exp.

Aux . Channels

Channel 5
Source VrB

Channel 6
Source VrA

Horn/Beacon lights / trailer control

External – plug into Rx Ch 6 (e.g. control light switch)

Aux . Channels

Channel 7
Source SwD

Channel 8
Source SwC

Virtual neutral (rev truck)

3 position Gear shift

Aux . Channels

Channel 9
Source SwB

Channel 10
Source None

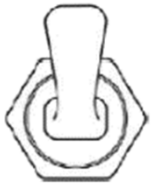
Un-couple 5th wheel servo (trailer control)

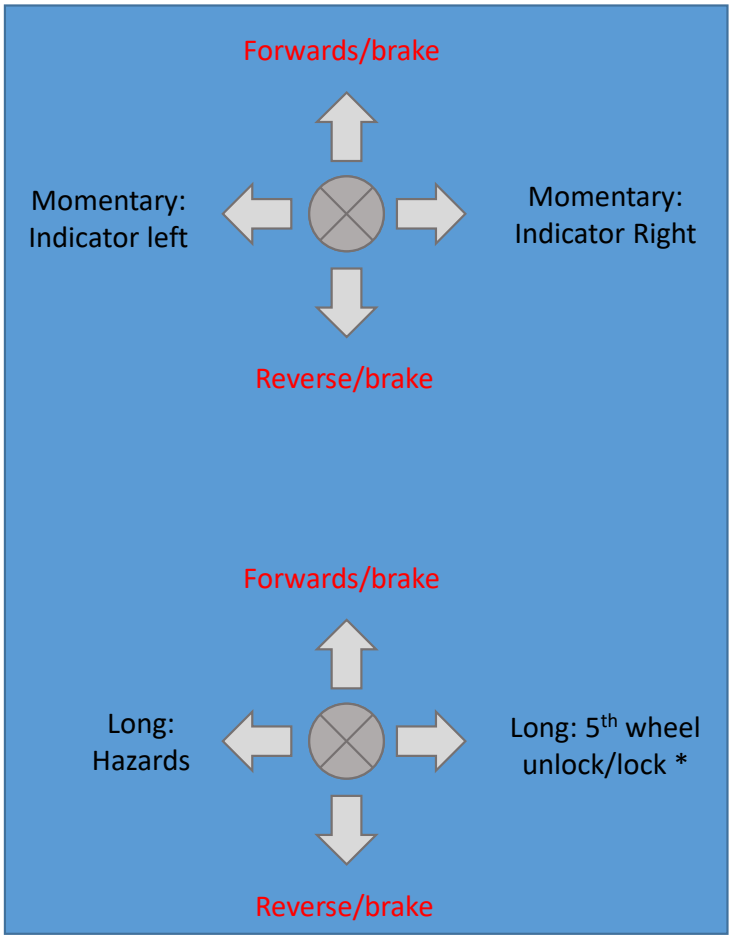
INTERNAL

ESP32 RC Sound & Light Controller

Remote Function L (left stick on remote)

Switch “SWA” 100%





Switch “SWA” 75%

*Long: 5th wheel unlock/lock

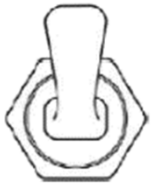
- Depending on setup
- Other option: SwB down

INTERNAL

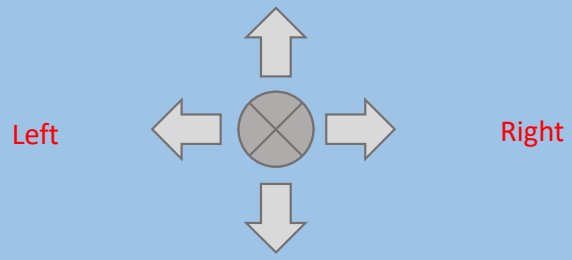
ESP32 RC Sound & Light Controller

Remote Function R (right stick on remote)

Switch “SWA” 100%



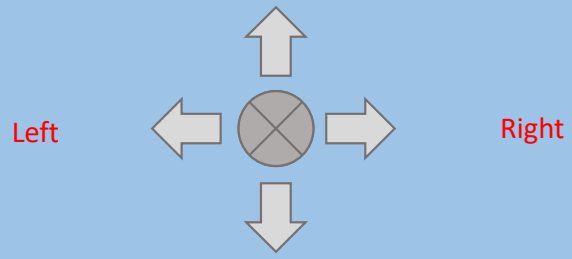
Lights off: momentary – headlight flash
Lights on: momentary – headlight flash
Lights on: Long – headlight on/off



Long - Idling: Volume (100, 66, 33%)
Long - Driving: Jake brake

Switch “SWA” 75%

Long – Switching through lights depending on settings in 6_adjustmentsLights.h



Long - Idling: Engine off

INTERNAL